NGO DANG HOANG VU

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Career Objective

Short-term goal: I aim to secure an opportunity in the field of VLSI, where I can apply the knowledge and skills I have gained through self-study and academic projects, while further developing my practical experience. I also hope to work in a professional and supportive environment that encourages continuous learning and growth.

Long-term goal: My goal is to gain valuable experience, continuously improve my technical expertise, and build a strong professional foundation. I aspire to grow within the organization and establish a long-term commitment.

Education

HCMC UNIVERSITY OF TECHNOLOGY AND ENGINEERING (HCMUTE) Ho Chi Minh City, Vietnam
Major: Computer Engineering *2022 – Present*

GPA: 7.62/10

Experience

TEST CHIP DESIGN - FSF0FS144A (Top-Chip Wire-Bond 55nm UMC)
Faraday Technology Vietnam - Internship Project *Sep 2025 - Dec 2025*

- Executed the complete Physical Design (PnR) flow using Cadence Innovus from Floorplan to postRoute for a Test Chip with 58K instances, 458 memory, 1.2V supply voltage and frequency 10MHz.
- Controlled the entire flow in Cadence Innovus: Analyzed results and debugged issues at each stage (Density map, Module Distribution, Congestion map, Timing, DRC/Connectivity/DRV, Power)
- Conducted Physical Verification with Calibre: Merged GDS files using Laker, then performed DRC/LVS checks and resolved all violations for prepare tape-out.
- Performed timing analysis and optimization: Worked on fixing setup/hold violations and collaborated with the ASIC Consultant team on timing constraints and clock structure.
- Performed midterm research on MOSFET/FinFET/GAA and Physical Cells; applied OCV-based STA in project flow and studied advanced STA concepts including AOCV/POCV and CPPR.

Projects

CUSTOM D FLIP-FLOP WITH INTEGRATED CLOCK GATING CELL
Timeline: Jan 2025 - Mar 2025

- Developed RTL models in SystemVerilog and verified their functional correctness via behavioral simulation.
- Designed and simulated both a standard D Flip-Flop and a clock-gated D Flip-Flop using latch-based clock gating technique in 90nm CMOS technology using Cadence Virtuoso.
- Implemented clock gating to reduce dynamic power consumption by suppressing unnecessary clock switching.
- Measured dynamic power consumption:
  - Standard DFF: 693.4 nW
  - AND-gated DFF: 479.2 nW
  - Latch-gated DFF: 381.5 nW

LAYOUT LOGIC GATES USING CMOS 90NM TECHNOLOGY

Timeline: Feb 2025 - Apr 2025

- Schematic Design: Created transistor-level designs for logic gates (NOT, NAND, NOR) at the transistor level using Cadence Virtuoso in 90nm CMOS technology.
- Layout Design & DRC/LVS: Designed physical layouts, ensured design rule compliance, and validated functionality using DRC & LVS.
- Skills Gained: CMOS – Cadence Virtuoso – Layout Design – DRC/LVS Verification.

DESIGN AND VERIFICATION TIMER IP

Timeline: Oct 2024 - Dec 2024

- Investigated and analyzed the requirements for the Timer IP.
- Designed the specification, including: APB protocol, 8 registers for different functionalities, and a counter.
- Wrote RTL code for the Timer IP by 2 module registers and counter.
- Reviewed RTL code for syntax, errors, and compliance with design requirements.
- Built the testing environment for RTL code verification.
- Using the full test cases to verify the entire RTL design.
- Result: 100%.

Technical Skills

Design Flow: Hands-on experience with Physical Design flow.
Static Timing Analysis (STA): Knowledge of timing constraints, setup/hold time.
Cmos Design: Understanding of basic digital circuit design (logic gates, simple circuits).
Clock Gating: Basic knowledge of clock gating techniques to reduce power consumption.
Digital Logic Design: Understanding of combinational and sequential circuits (flip-flops, latches, mux, counter, FIFO, ALU...)
Communication Protocols: Understanding of APB Protocol.
Hardware Design Language: Basic proficiency in Verilog/SystemVerilog.
Programming Language: Basic proficiency in C/C++ and Tcl, Python scripting.
EDA Tools: Cadence Innovus, Cadence Virtuoso, Calibre, Laker, Xilinx Vivado, Quartus, ModelSim, Matlab, Kicad.
Operating System: Linux (Ubuntu, CentOS).
Tools: Git, GitHub, Docker, Vim Editor, Wavedrom.

Soft Skills

Individual work: Self-study, hard-working, task management, problem solving and analysis.
Team work: Responsibility, communication, time management.

Languages

Vietnamese: Native
English: Professional proficiency (TOEIC 620)

Certificates

FARADAY TECHNOLOGY VIETNAM (FTV) : Physical Design Intern
UDEMY: VSD - Physical design flow